ClimateAdapt Pune

End-to-End Implementation Guide

From climate data to climate action

Detailed project, data, analytics, optimization, dashboard and reporting specification

Pilot: Pune, Maharashtra, India
Prepared: 17 October 2026

How to use this guide

This is a complete working specification, not a claim that every feature has already been implemented or every dataset has been downloaded. Work through the phases in order, preserve original files, and only activate analyses when the required data and metadata are available.

Project status reflected here: OSM GeoJSON layers and some OpenCity drainage resources have been reported downloaded; an official 2011 Census workbook and Maharashtra groundwater resource report have been located/downloaded; the Open-Meteo API test succeeded. Sentinel-2 and DEM downloads, local flood records and local groundwater monitoring data are not confirmed.

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1. Project overview

Purpose

ClimateAdapt Pune is an AI-assisted climate adaptation action planner for a defined pilot area in Pune. It combines climate, geographic, infrastructure, environmental and population data to identify potential vulnerability, explain the evidence behind it, recommend practical interventions, compare plans under a budget, and produce a transparent report and dashboard.

Core principle

The system must distinguish measured or officially reported facts from model forecasts, proxies, assumptions and missing data. A map layer being present does not prove that infrastructure is functional; absence of complaint records does not mean there were no floods; a model estimate is not a field measurement.

Pilot scope

Start with the current Pune pilot bounding box (approximately 18.50–18.60 latitude and 73.80–73.90 longitude). Keep the geographic extent configurable so the system can later support another ward, campus, housing society or city.

Primary users

Municipal planners, ward officers, campus or society administrators, climate-resilience teams, researchers and community organizations.

Main outputs

A data-quality summary, vulnerability indicators with evidence, an interactive map, ranked interventions, a budget-aware action plan, scenario comparison, downloadable report, and a record of assumptions and data freshness.

2. Feature specification

A. Geographic project setup

Allow a user to select a study area using a ward boundary, uploaded GeoJSON, or a configured bounding box. Show the study-area boundary, coordinate reference system, area, source and last-updated date. Validate geometry, repair invalid polygons where safe, and clip spatial layers to the study area.

B. Data-source and freshness panel

For each layer show source, download/retrieval date, period covered, geographic resolution, units, license/attribution, processing status and confidence. Flag stale, incomplete, mismatched or unavailable data. Provide a source register and data dictionary.

C. Heat exposure and heat-vulnerability analysis

Use temperature, apparent temperature, humidity, land-cover/vegetation, built-up context and population where compatible data exist. Summarize hot periods and seasonal patterns; map heat-related proxies such as low vegetation cover and high built-up density. Keep exposure separate from sensitivity and adaptive capacity. Do not label the result as a validated heat-risk prediction without validation data.

D. Flood and intense-rainfall screening

Use rainfall totals/probability, elevation/terrain derivatives when available, waterways, drainage infrastructure, impervious/built-up proxies and verified historical flood reports if available. Show low-lying areas and drainage proximity as screening indicators, not proof of flooding. Mark complaint history and drain condition as missing until supported by records.

E. Water-stress and drought screening

Use rainfall history, evapotranspiration, soil-moisture model estimates, groundwater resource reports and verified water-supply information. Distinguish district/taluka resource assessments from local monitoring-well measurements. Where local groundwater levels are absent, present regional context and clearly label local groundwater status as unavailable.

F. Green-cover and nature-based resilience

Use Sentinel-2 imagery when downloaded and appropriate cloud-free coverage is confirmed. Derive NDVI from B08 (near-infrared) and B04 (red) bands using (B08 - B04) / (B08 + B04). Map vegetation/green-space proxies and OSM parks, gardens, grass and woodlands. Explain that NDVI is a vegetation signal, not a direct measure of tree health, shade quality or public accessibility.

G. Population and vulnerable groups

Use official census or other authoritative population data only when the geographic units match or can be defensibly crosswalked. Show total population and available age-group indicators. Never divide city population equally among wards or assume historic census boundaries equal current ward boundaries. If a defensible match is not possible, mark the statistic unavailable.

H. Infrastructure and essential services

Map OSM buildings, roads, schools, hospitals, clinics, drinking-water points, waterways, water bodies, parks and green spaces. Include downloaded official/OpenCity stormwater-drain and coverage datasets after inspecting file format, metadata and fields. Presence of a drain line or coverage polygon does not establish condition, capacity, blockage status or actual service performance.

I. Evidence-backed explanations

Every indicator and recommendation should show a human-readable explanation, the input layers used, dates and resolution, known gaps, and whether it is an observation, proxy, forecast or assumption. Let users inspect evidence rather than presenting an unexplained single score.

J. Intervention catalogue

Provide editable interventions such as shade trees/corridors, cool roofs, shaded bus stops/walking routes, rainwater harvesting, permeable surfaces, rain gardens, drainage inspection and maintenance, stormwater inlet clearing, detention/retention areas, restoration of waterways, water conservation, drought contingency planning and protection/restoration of green spaces. Each entry should have applicability conditions, expected benefits, estimated cost range, lead time, maintenance needs, responsible stakeholder and evidence source.

K. Budget-aware action planning

Let users set a budget, time horizon, constraints, intervention eligibility and priorities. Rank interventions using transparent weighted criteria such as risk reduction, population benefit, equity, cost, feasibility, co-benefits and maintenance. Provide a simple baseline ranking first; optionally use an optimization solver to choose a portfolio under budget and constraints. Display the rationale and trade-offs.

L. Scenario simulator

Allow users to change assumptions such as a heatwave, heavy-rainfall scenario, vegetation improvement, water-supply restriction or budget change. Compare baseline and proposed actions using clearly labelled proxy indicators. Never imply a physically validated impact simulation unless calibrated models and suitable local data exist.

M. Equity and fairness

Show which neighborhoods or sectors benefit, where data are missing, and whether the plan directs resources only to areas with better data. If population or vulnerability data are incomplete, surface that limitation. Provide configurable equity weights and explain how they affect rankings.

N. Interactive dashboard and map

Provide overview cards, time-series charts, layer toggles, a map legend, filters, data-source details, ward/sector drill-down, before/after scenario comparison and an evidence panel. Include accessible colors, readable labels, mobile-friendly layout and a clear explanation of uncertainty.

O. Report generation and exports

Generate a downloadable PDF and CSV/GeoJSON exports where relevant. The report should include study-area scope, date, sources, methods, data-quality notes, indicators, maps/charts, prioritized actions, costs, assumptions, uncertainty, limitations and next data-collection steps.

P. Administration and auditability

Keep configuration separate from code; log ingestion time and processing outcomes; version indicator definitions and action-plan runs; validate uploaded files; protect secrets in environment variables; avoid exposing private data; and retain enough metadata to reproduce a result.

3. Data inventory and current collection status

Ward boundary

Preferred source: official municipal ward boundaries or a verified boundary file. If unavailable, the pilot bounding box can be used temporarily and must be labelled as a study extent, not an official ward boundary.

OpenStreetMap

Files already reported saved under data/raw/osm/: parks.geojson, buildings.geojson, roads.geojson, trees.geojson, schools.geojson, hospitals.geojson, waterways.geojson, water_bodies.geojson, clinics.geojson, gardens.geojson, grass.geojson, woodlands.geojson, drinking_water.geojson. Inspect feature counts, geometry types, timestamps if available and coverage before analysis.

Drainage

Stormwater-drain resources and ward-level drainage CSVs were reported downloaded from OpenCity. Inspect actual filenames, formats, columns, units, spatial coverage, provenance and license before integrating. Do not assume all downloaded resources are valid GeoJSON; KML/KMZ or tabular files may need conversion.

Weather

Open-Meteo Forecast API test succeeded. The response contained current, hourly and daily variables for coordinates 18.523726, 73.86876, including temperature, relative humidity, apparent temperature, precipitation, cloud cover, wind, evapotranspiration, ET0, soil moisture estimates and UV index. The initial generated request used a different coordinate pair; use a documented, intentional pilot point and keep its coordinates in metadata. Forecast data must remain separate from historical climate data.

Population

An official Census Digital Library 2011 Primary Census Abstract file was found for Pune district. Before use, check sheet names, geographic level and whether its ward/village/town units align with the current study area. Do not infer current pilot population from a mismatched geography.

Groundwater

A Maharashtra Groundwater Resource Estimation report for 2024 was downloaded, and a Pune row was located. Interpret values only after confirming the exact column headers and geographic level. Regional resource estimates are not equivalent to current groundwater levels in local wells.

Sentinel-2

A 15 October 2026 Sentinel-2 L2A product was identified, with product-level cloud cover reported around 8.1%. It is not confirmed that the product was downloaded or that the pilot area is cloud-free. Inspect pilot-area coverage before calculating NDVI.

DEM/elevation

No confirmed DEM download is recorded. Obtain a suitable elevation raster from a credible source, document resolution and vertical datum, and derive slope/flow-related indicators only after validating the raster.

Missing or unverified

Historical flood complaint records, local drain blockage/condition, local groundwater-monitoring time series, and some OSM layers (such as drains, wells, community centres and transport stops) have not been confirmed. Treat these as missing, not zero.

4. Recommended folder structure

Project layout

```
climateadapt-pune/
  data/
    raw/
      wards/
      weather/
      osm/
      sentinel2/
      dem/
      population/
      official_reports/
    processed/
    exports/
  backend/
    app/
      main.py
      database.py
      models.py
      schemas.py
    routes/
  scripts/
    fetch_weather.py
    download_osm_notes.md
    validate_data.py
    process_indicators.py
    build_action_plan.py
    generate_report.py
  frontend/
  reports/
  requirements.txt
  .env
  README.md
```

Data handling rule

Keep downloaded originals unchanged under data/raw/. Save cleaned or transformed datasets under data/processed/. Save user-facing tables, maps and reports under data/exports/ or reports/. Never silently overwrite raw source files.

5. End-to-end implementation workflow

Phase 1 — Establish the study area

Confirm the bounding box or official boundary, coordinate system and pilot metadata. Add a project configuration file for area name, extent, timezone, source locations and run date. Validate that all later layers are clipped to the same study area.

Phase 2 — Inventory and validate raw files

Create a data manifest listing filename, type, source, retrieval date, coverage, CRS, feature/row count and status. Run scripts/validate_data.py to check JSON/GeoJSON validity, CSV/XLSX/PDF presence, empty files, invalid geometries, duplicate records, missing fields, coordinate ranges and obvious unit problems. Report issues instead of silently deleting data.

Phase 3 — Fetch weather

Use scripts/fetch_weather.py to request Open-Meteo data with requests. Save the complete response as data/raw/weather/pune_forecast_raw.json, plus hourly and daily CSVs. Include retrieval timestamp, coordinates, timezone and data_type='forecast'. Handle HTTP failures, timeouts and malformed responses. Do not describe forecast output as historical observations.

Phase 4 — Inspect OSM and official infrastructure data

Load each GeoJSON and report counts and geometry types. Inspect OpenCity stormwater resources and drainage CSV headers before transforming. Reproject layers to a suitable projected CRS for area/distance calculations; retain source CRS metadata. Avoid using latitude/longitude degrees for planar area or distance calculations.

Phase 5 — Acquire historical climate and environmental layers

Fetch historical weather separately from a suitable historical API/source and define a meaningful time window (for example, multiple years where available). Record data provenance and whether values are reanalysis/modelled or station observations. Inspect Sentinel-2 image coverage and clouds; derive NDVI only after downloading the bands. Obtain a suitable DEM if terrain-based flood screening is in scope.

Phase 6 — Population and groundwater validation

Inspect Census workbook sheets, geography and columns; map records only where a reliable boundary correspondence exists. Inspect groundwater report headings and geographic unit before parsing the Pune row. Record unavailable local measures as missing.

Phase 7 — Clean and standardize

Standardize dates to ISO format, units, column names, null handling and CRS. Deduplicate records where appropriate. Preserve raw values and add provenance fields. Distinguish true zero from null/missing. Create a data-quality report.

Phase 8 — Database and spatial storage

For a small prototype, GeoJSON/CSV files may be sufficient initially. For multi-layer queries and a production-style app, use PostgreSQL with PostGIS. Create tables for study areas, data sources, weather observations/forecasts, infrastructure features, population statistics, indicators, interventions, action-plan runs and audit metadata. Add spatial indexes and foreign keys. Use migrations or repeatable SQL scripts.

Phase 9 — Calculate indicators

Implement reproducible indicator functions in scripts/process_indicators.py. Store formula, input layer versions, spatial unit, time window, normalization method, output units and limitations with every indicator. Use min-max or percentile scaling only when appropriate and document the reference population. Do not mix incomparable scales without normalization.

Phase 10 — Build the action plan

In scripts/build_action_plan.py, combine evidence-backed vulnerability indicators with intervention eligibility, cost estimates and benefit assumptions. Start with a transparent weighted ranking. Add budget optimization only after inputs

and constraints are defined. Generate a machine-readable action-plan JSON/CSV containing scores, reasons, costs, dependencies and confidence.

Phase 11 — Generate a mini-report early

Use scripts/generate_report.py to create a small evidence-led Pune report before building the full dashboard. Include current data inventory, sample maps/charts, a few indicators, candidate interventions and missing-data warnings. This catches data issues early.

Phase 12 — Backend API

Use FastAPI or the framework already selected in the project. Create endpoints for project metadata, data-source status, indicators, map features, interventions, action-plan runs, scenario comparisons and report downloads. Validate request/response schemas, handle errors and document the API.

Phase 13 — Frontend dashboard

Build a React dashboard if that is the chosen frontend. Create a map view, overview page, vulnerability detail, intervention planner, scenario comparison and report/export area. Fetch all displayed values from the API and label the source/time window.

Phase 14 — Test, document and demonstrate

Test data ingestion, spatial alignment, missing-data handling, calculations, budget constraints, API responses, exports and dashboard states. Prepare a demo scenario, sample report, README, architecture diagram and known-limitations list.

6. Indicator framework

Heat screening

Candidate inputs: maximum temperature/apparent temperature, frequency of hot days if historical data support it, vegetation/NDVI, built-up density, shade proxies and population exposure. Potential output: heat exposure score and separate sensitivity/adaptive-capacity indicators. Do not infer individual health outcomes.

Flood screening

Candidate inputs: rainfall intensity/totals, DEM-derived low-lying areas and slope, waterways, drainage proximity/coverage and verified flood history. Potential output: screening priority, not a calibrated flood probability unless a validated hydrologic/hydraulic model exists.

Water stress

Candidate inputs: historical precipitation deficit, ET0, soil moisture estimates, verified water supply information and groundwater resource context. Potential output: water-stress screening with a clear distinction between regional and local evidence.

Green-cover deficit

Candidate inputs: NDVI, land cover, mapped parks/gardens/woodlands and population distribution. Potential output: green-cover proxy or access-gap estimate. Do not claim public accessibility without access/entrance data.

Equity

Where population or vulnerability indicators are reliable, compare who benefits and whether high-need areas are served. Keep a visible warning where demographic data are old, coarse or mismatched.

Composite score

If a composite score is used, publish the formula, weights, normalization, missing-data policy and sensitivity to weights. Provide component scores beside the composite. Avoid false precision such as reporting several decimal places for weak proxies.

7. Intervention catalogue and planning logic

Heat actions

Tree/shade corridors; cool-roof pilots; shaded public waiting areas; shade for walking routes; protection of existing tree canopy; heat-risk communication. Check water availability, species suitability, utility conflicts, maintenance and land ownership.

Flood actions

Drain inspection/maintenance; inlet cleaning; rain gardens; permeable surfaces; detention/retention areas; restoration of natural drainage corridors; targeted drainage upgrades after engineering review. Do not recommend construction based only on proximity or a coarse map.

Water actions

Rainwater harvesting; leak detection and repair; water-use efficiency; drought contingency plans; groundwater monitoring; recharge measures only after hydrogeological assessment; protection of local water bodies.

Green/nature actions

Protect existing green spaces; native planting; restore degraded patches; connect green corridors; create small shade/rest areas. Track maintenance and survival rather than counting planting events alone.

Ranking criteria

Risk reduction, population served, equity, cost, feasibility, implementation time, co-benefits, maintenance burden, evidence strength and reversibility. Weights should be configurable and visible.

Budget optimization

Decision variable x_i indicates whether intervention i is selected (or its scale). A simple formulation maximizes total weighted benefit subject to sum(cost_i × x_i) <= budget and eligibility/dependency constraints. Benefit and cost estimates must be labelled as estimates and should be adjustable.

Scenario comparison

Compare baseline and intervention portfolio under the same assumptions. Display change in proxy indicators, budget used, people/areas potentially served, confidence and unresolved risks. Avoid presenting proxy deltas as guaranteed real-world outcomes.

8. Database and API design

Suggested entities

Project/StudyArea; DataSource; DataAsset; WeatherRecord; SpatialFeature; PopulationStatistic; IndicatorDefinition; IndicatorResult; Intervention; InterventionCostAssumption; ActionPlan; ActionPlanItem; Scenario; ReportArtifact; ProcessingRun.

Important fields

Each data asset should include source_url, provider, retrieved_at, period_start/end, geometry/CRS, resolution, license, data_type, processing_version and quality_status. Indicator results should include formula_version, spatial_unit, input_asset_ids, value, unit, confidence/quality label and limitations.

API endpoints

GET /health; GET /api/project; GET /api/data-sources; GET /api/indicators; GET /api/map-layers; GET /api/interventions; POST /api/action-plans; POST /api/scenarios/compare; GET /api/reports; GET /api/reports/{id}/download. Adapt prefixes and conventions to the existing app.

Security

Keep credentials in .env and never commit secrets. Validate uploads and file types. Limit file sizes, use parameterized SQL, set CORS deliberately, and avoid exposing private information or internal file paths.

9. Dashboard design

Overview

Study area, last data refresh, source completeness, major screening indicators, budget and action-plan status. Every card should show period, units and source or link to details.

Map explorer

Toggle heat, flood-screening, water-stress, green-cover and infrastructure layers. Include legend, opacity controls, click-to-inspect features and layer metadata. Use clear labels for unavailable or proxy layers.

Risk details

Show component scores, underlying evidence, data resolution, known limitations and missing-data warnings. Let users compare neighborhoods only when spatial units and inputs are comparable.

Action planner

Budget control, priority weights, time horizon, eligible interventions and constraints. Display selected actions, estimated costs, expected proxy benefits, equity distribution and why each item was selected.

Scenario studio

Change assumptions and compare baseline with alternatives. Include side-by-side indicators and a clear statement of which parts are modelled, assumed or observed.

Reports and exports

Generate PDF, CSV and appropriate spatial exports. Include the exact run timestamp and configuration so a result can be reproduced.

10. PDF report contents

Recommended sections

Title and study area; executive summary; scope and methodology; data inventory and source dates; data-quality and coverage map; heat screening; flood screening; water-stress screening; green-cover analysis; population/equity assessment; prioritized interventions; budget and scenario comparison; implementation roadmap; limitations and uncertainty; missing-data collection plan; sources and attribution.

Visuals

Study-area map, layer coverage, temperature time series, rainfall chart, NDVI/green-cover map where available, infrastructure map, indicator comparison chart, action portfolio and budget breakdown.

Reproducibility

Report should include project configuration, processing run ID, indicator formula versions, data timestamps, intervention cost assumptions and source links.

11. Testing and acceptance criteria

Ingestion tests

A valid source downloads and parses; failed requests produce useful errors; original data are retained; timestamps and metadata are recorded.

Geospatial tests

CRS is recognized; invalid geometries are reported; layers align spatially; area/distance calculations use a projected CRS; data are clipped to the pilot area.

Data-quality tests

Nulls are not silently converted to zero; duplicates and unexpected units are flagged; forecast and historical data are separated; geographic mismatch is detected.

Indicator tests

Known sample inputs produce expected outputs; formulas and normalization are versioned; missing input handling is deterministic; component scores are inspectable.

Planner tests

Selected interventions do not exceed the budget; dependencies and eligibility are respected; rankings respond predictably to weight changes; cost assumptions are visible.

Report/API/UI tests

Report exports open correctly; API schemas are validated; errors are handled; maps show legends; loading/empty/error states are clear; accessibility and basic responsive layout are checked.

Acceptance definition

A demonstrable MVP should ingest and validate real sourced data, show at least one reliable indicator and one explicitly labelled proxy, explain its evidence, produce a constrained action plan, compare a scenario, and generate a report without fabricating missing values.

12. Responsible use, uncertainty and known limitations

No invented data

Never create fake drain conditions, flood events, groundwater levels, ward populations or vegetation values to fill gaps. Use null/missing and explain what data would resolve the gap.

No overclaiming

A forecast is not an observation. A proxy score is not a validated probability. Regional groundwater estimates are not local well measurements. OSM completeness varies. Satellite cloud cover must be checked over the study area.

Uncertainty

Use qualitative confidence labels tied to source quality, freshness, spatial resolution and validation. If a numeric confidence score is used, define how it is calculated and validate it.

Human review

Engineering decisions such as drainage upgrades, recharge interventions or construction require qualified local assessment and approvals. The tool supports planning; it does not replace official warnings or engineering design.

Privacy and equity

Use aggregated population data where possible. Do not expose household-level sensitive information. Track which communities are underrepresented because of missing data and allow planners to flag those gaps.

13. Suggested development order for the current project

First

Inspect and validate the data files already collected. Keep the current folder structure.

Second

Make scripts/fetch_weather.py save Open-Meteo raw JSON plus hourly/daily CSVs, including source and retrieval metadata.

Third

Inspect OpenCity stormwater resources and the Census workbook, recording actual filenames, formats and column headers.

Fourth

Confirm whether Sentinel-2 and a DEM have actually been downloaded; do not treat identified products as downloaded assets.

Fifth

Build validate_data.py and a machine-readable data manifest.

Sixth

Create a small first-pass indicator pipeline with only supported inputs and explicit missing-data handling.

Seventh

Generate an early mini-report before investing time in a complex dashboard.

Eighth

Build action ranking and budget constraints; add scenario comparisons after baseline calculations are tested.

Ninth

Build backend and React dashboard against stable API responses.

Tenth

Run acceptance tests, document limitations and prepare the demo.

14. Suggested README and demo narrative

README essentials

Project objective; setup instructions; environment variables; folder structure; data source register; how to run ingestion/validation/processing/report scripts; how to launch backend/frontend; known limitations; license/attribution; test commands.

Demo flow

Select Pune pilot area → inspect source completeness → open a climate screening layer → inspect why a location is flagged → choose a budget and priorities → compare an intervention portfolio → view equity and uncertainty → export the report.

Honest demo language

Say 'screening indicator' or 'proxy' when appropriate. Explain which datasets are available and which are missing. Do not claim the system predicts exact flood events or proves an intervention's impact unless validated.

15. Reference sources and attribution

Open-Meteo

Forecast API: https://api.open-meteo.com/v1/forecast. Document API parameters, retrieval time, coordinates and forecast nature.

OpenStreetMap

Data: https://www.openstreetmap.org/ and Overpass Turbo: https://overpass-turbo.eu/. Follow OpenStreetMap attribution and applicable ODbL requirements.

OpenCity Pune stormwater data

https://data.opencity.in/dataset/pune-stormwater-drains and https://data.opencity.in/dataset/?tags=City%20Services&groups=pune. Verify individual resource metadata, license and update date.

Census of India

Pune district 2011 Primary Census Abstract record: https://censusindia.gov.in/nada/index.php/catalog/6712. Verify geography and workbook fields before analysis.

GSDA Maharashtra

Groundwater resources and monitoring information: https://gsda.maharashtra.gov.in/en-dashboard/ and https://gsda.maharashtra.gov.in/en-national-hydrology-project-2/. Confirm report headers and spatial level.

India-WRIS

https://indiawris.gov.in/wris/. Check availability and metadata for relevant hydrology data.

Copernicus Browser

https://browser.dataspace.copernicus.eu/. Verify product footprint, tile, sensing date and cloud cover over the actual pilot area.

Bhuvan / NRSC

https://bhuvan.nrsc.gov.in/ and DEM help: https://bhuvan-app3.nrsc.gov.in/data/download/help/source/html/introduction.htm. Verify download terms and raster metadata.